Shahkar Ahmad Shah

AI Engineer · LangGraph & Agentic AI Specialist · LLM Systems · Python

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PROFESSIONAL SUMMARY

Final-year Computer Science student and self-driven AI Engineer with hands-on experience building production-grade Agentic AI systems using LangGraph, MCP, and Python. Designed and shipped a stateful multi-agent pipeline for autonomous content generation — spanning research, writing, fact-checking, podcast synthesis, and video production. Strong ML/DL foundations including CNNs, MLPs, and predictive modelling. Passionate about understanding how models work under the hood, not just consuming APIs. Open to remote AI Engineer roles and international freelance contracts.

TECHNICAL SKILLS

Agentic AI	LangGraph · LangChain · MCP · Multi-Agent Orchestration · Human-in-the-Loop · RAG Pipelines
LLMs & GenAI	GPT-4o · Gemini 2.5 Flash · OpenAI API · Prompt Engineering · Tavily Search Integration
ML / DL	CNN · MLP · Perceptron · Scikit-learn · Pandas · NumPy · Predictive Modelling
Backend	FastAPI · SQLite · Streamlit · REST APIs
Tools & Core	Python (Advanced) · PyTorch · TensorFlow · Git · GitHub · Jupyter
Languages	English · Urdu · Pashto

PROJECTS

Autonomous Multi-Agent Content Ecosystem · Final Year Project (Ongoing) *2024 – 2025*

Stack: LangGraph · GPT-4o · Gemini 2.5 Flash · Tavily · FastAPI · Python 3.11 · MoviePy

- Architected a complex stateful **LangGraph workflow** coordinating 5+ specialised autonomous agents (Router, Researcher, Orchestrator, Writer, Auditor) to produce enterprise-grade long-form content from a single topic prompt
- Implemented parallel fan-out agent execution — dispatching concurrent Worker agents per blog section — reducing generation time significantly vs. sequential processing
- Built a **Human-in-the-Loop** (HITL) memory persistence layer using LangGraph interrupts, allowing users to pause execution, review the strategic plan, and inject edits before costly generation begins
- Integrated a Fact-Checker Agent that cross-references all AI-generated claims against Tavily-scraped source data, producing a quantifiable quality score to prevent hallucinations
- Engineered a native AI Podcast Studio using Gemini 2.5 Flash audio API to synthesise 3-minute conversational podcasts from generated blog content
- Added automated Video Generator pipeline using MoviePy, Pexels B-roll API, TTS voiceover, and subtitle overlay — producing ready-to-upload short-form social video

ODI Run Predictor — Full-Stack ML App · Personal Project *2024*

Stack: FastAPI · Streamlit · Scikit-learn · Python

- Trained a machine learning model achieving 92% accuracy (R^2 score) and MSE of 0.11 to predict cricket ODI scores from match conditions
- Exposed the model via a FastAPI REST endpoint so external applications can consume predictions programmatically
- Built an interactive Streamlit frontend for non-technical users to run predictions in real time

YouTube Video Summariser · Personal Project *2024*

Stack: LangChain · YouTube Transcript API · GPT-4 · Python

- Built an LLM-powered pipeline that extracts YouTube transcripts and generates structured chapter-by-chapter summaries using LangChain and GPT-4

- Supports multiple output formats: bullet summaries, key insights, and action items tailored to content type

Intra Campaign Monitoring — Polio Drive Analytics · Real-World Impact Project *2024*

Stack: Python · Pandas · Scikit-learn · Streamlit

- Designed a predictive system to forecast 'Not Available Children' reports during vaccination drives, directly supporting field managers in resource planning
- Cleaned and processed real-world messy datasets — handling missing values, inconsistencies, and outliers — to achieve a model accuracy of 91%
- Delivered an interactive Streamlit dashboard for field supervisors to monitor predictions and campaign coverage in real time

EDUCATION

BS Computer Science — 7th Semester · University of Peshawar, Pakistan *2021 – 2026*

- Final Year Project: Autonomous Multi-Agent AI Content Ecosystem (LangGraph · Agentic AI · LLMs)
- Relevant coursework: Machine Learning · Deep Learning · Data Structures · Software Engineering · Database Systems

Open to remote AI Engineer roles · Freelance LLM / Agentic AI contracts · International teams · UTC +5 (Flexible)